Jay Doshi

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Education

B.Sc. in Computer Science & Engineering **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

Professional Projects

DeepTruth-AI | Python, PyTorch, DistilBERT, AWS, GCP, Gemini API, MongoDB, React **January 2025 – March 2025**

- Automated scheduled Transformer retraining and ML lifecycle management with AWS Lambda and CloudWatch cron workflows.
- Cleaned 9,500+ records with a Python ETL pipeline and automated validation checks, improving data ingestion accuracy.
- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Streamlined model deployment with CI/CD via GitHub Actions and GCP VMs, reducing manual release steps.

SkillBridge Learning Platform-AI | NLP, TF-IDF, PostgreSQL, Django, AWS **August 2024 – December 2024**

- Built an NLP recommendation engine using TF-IDF and cosine similarity, achieving 95% prediction accuracy in testing.
- Designed PostgreSQL schemas with fact and dimensional modeling, improving query performance and reporting speed by 80%.
- Developed a Gemini API-powered AI agent auto-generating video transcripts and MCQs, reducing manual content creation effort.

SportsSight Analytics-AI | Python, TensorFlow, YOLOv8, OpenCV, K-Means, NumPy, Matplotlib **January 2025 – April 2025**

- Fine-tuned YOLOv8 on Roboflow football dataset, achieving 90% player and ball detection accuracy on test video clips.
- Applied K-Means pixel clustering for automated team assignment by t-shirt color, improving segmentation accuracy to 75%.
- Generated real-time match analytics on ball possession and movement patterns to support data-driven sports insights.

PolicyRAG Document Intelligence-AI | Python, RAG, GPT-4o, FAISS, FastAPI, Streamlit **December 2025 – January 2026**

- Built a RAG pipeline over 5 insurance policy documents (500+ pages), parsing unstructured PDF text and tables with pdfplumber.
- Indexed chunks in a FAISS vector store using OpenAI embeddings, reranking by cosine similarity to feed GPT-4o the top 5 clauses.
- Deployed a FastAPI endpoint with token auth answering batched queries in 40–50s; benchmarked retrieval latency and accuracy.

Experience

Network Engineer Intern | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Monitored network and server health in SolarWinds NPM, flagging performance degradation before it escalated into outages.
- Collaborated with infrastructure, OT, and process-control engineers to validate system integrations and support production cutovers.

Data Processing Engineer Intern | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Automated ETL ingestion and transformation of OpenVMS Alpha sensor data into SQL Server, cutting manual extraction by 40%.
- Engineered a classification-based anomaly detection tool analyzing live gauge data, reducing defective steel plate escapes by 30%.
- Developed Power BI semantic models on SQL Server views in a star schema, validating totals and accelerating decisions by 50%.

IT Support Technician Level 2 | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Managed 200+ IT assets through inventory tracking, device deployment, secure decommissioning, and life-cycle documentation.

Undergraduate Research Assistant | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Increased control accuracy by 60% via statistical analysis of RMSE and overshoot in Pandas across 15+ hardware trials.
- Designed and evaluated a PPO-based reinforcement learning model for autonomous drone target navigation using ROS2-Gazebo.
- Decreased target navigation error by 40% by tuning a custom PID-based flight model in a ROS2-Gazebo digital twin.

Technical Skills

Programming Languages & Frameworks: Python, SQL, R, C++, Java, JavaScript, FastAPI, Django, React

AI & Machine Learning: Supervised/Unsupervised Learning, Model Evaluation, NLP, MLOps, Fine-Tuning, Transformer Models, Deep Learning, Reinforcement Learning, LLM, RAG

Data & Visualization: Pandas, NumPy, Matplotlib, Power BI Dashboards, Tableau, Data Cleaning, Data Modeling, Excel

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS (Lambda, S3, EC2), Google Cloud Platform, Microsoft Azure

Developer Tools & ML Libraries: Git, GitHub Actions, PyTorch, TensorFlow, Hugging Face, Jupyter Notebook, Google Colab, Agile/SCRUM